

Class activity: linear regression and transformations

Group members:

Instructions: Work with a neighbor on the following activity. I will collect the handout at the end of class, and it will be part of your class participation grade. You will be graded only on effort – it is ok if you don't finish all the questions, or get them all correct.

GDP and life expectancy

Consider the fitted model

$$\widehat{\text{LifeExpectancy}} = 4.95 + 7.20 \log(\text{GDP per capita})$$

For a GDP per capita of \$20000, we calculated a predicted life expectancy of 76.255 years.

1. Consider a 1% increase in GDP per capita – from \$20000 to \$20200. What is the new predicted life expectancy from a GDP per capita of \$20200? (*Remember that here log means natural log!*)
2. What is the difference in the estimated predictions? How does this difference compare to the estimate slope $\hat{\beta}_1 = 7.20$?

Housing prices

Consider the fitted model

$$\log(\widehat{\text{Price}}) = 10.51 + 0.091 \text{ Area}$$

where price is in dollars, and area in hundreds of square feet.

For a house of 2000 square feet (i.e. area = 20), we calculated a predicted price of \$226386.

3. Calculate the predicted price for a house of 2100 square feet.

4. Calculate the *ratio* between your prediction in question 3, and the predicted price for a house of 2000 square feet.

5. Now calculate $e^{0.091}$. How does this compare to your answer in question 4?